

3-b Family Facts in Prolog

PROGRAM:

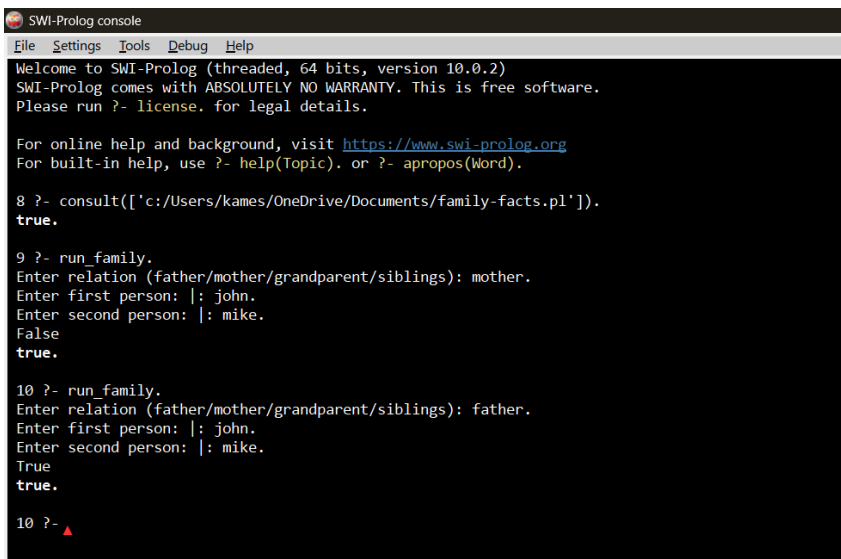
```
% Facts
parent(john, mike).
parent(john, susan).
parent(mary, mike).
parent(mary, susan).

male(john).
male(mike).
female(mary).
female(susan).

% Rules
father(X, Y) :- parent(X, Y), male(X).
mother(X, Y) :- parent(X, Y), female(X).
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
siblings(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.

% Interactive query
run_family :-
    write('Enter relation (father/mother/grandparent/siblings): '), read(Rel),
    write('Enter first person: '), read(P1),
    write('Enter second person: '), read(P2),
    (call(Rel, P1, P2) -> writeln('True') ; writeln('False')).
```

OUTPUT:

A screenshot of the SWI-Prolog console window. The window has a title bar "SWI-Prolog console" and a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The console text shows the welcome message, version 10.0.2, and a license notice. It then shows the execution of a Prolog program. The first query is to consult a file, which succeeds. The second query is to run the family program, which prompts for a relation, first person, and second person. The first run with relation "mother", first person "john", and second person "mike" returns "False". The second run with relation "father", first person "john", and second person "mike" returns "True". The prompt "10 ?-" is shown at the bottom with a red cursor.